TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	93.9 ± 0.54
Correlation Stability	95.7 ± 0.64
Distribution Consistency	96.1 ± 0.54
Entropy Preservation	97.6 ± 0.49
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	94.3 ± 2.10
Feature Accuracy	97.6 ± 0.49
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	4.081	+0.236	6.1%	OK
Glucose	120.895	122.552	+1.658	1.4%	OK
BloodPressure	69.105	69.142	+0.036	0.1%	OK
SkinThickness	20.536	20.979	+0.442	2.2%	OK
Insulin	79.799	79.091	-0.708	0.9%	OK
BMI	31.993	31.877	-0.115	0.4%	OK
DiabetesPedigreeFunction	0.472	0.466	-0.006	1.2%	OK
Age	33.241	33.619	+0.378	1.1%	OK
Outcome	0.349	0.363	+0.014	4.1%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	93.9 ± 0.54
Correlation Stability (*2)	95.7 ± 0.64
Distribution Consistency (*3)	96.1 ± 0.54
Entropy Preservation (*4)	97.6 ± 0.49
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	94.3 ± 2.10
Feature Accuracy (*10)	97.6 ± 0.49

Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.729220779220779 2	0.712987012987013	0.729220779220779 2	-1.1102230246251566e -17	
F1	0.592152063889281	0.550731192512098 3	0.591213623316142 2	-0.0009384405731386103	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.754545454545454 5	0.746103896103896	0.755194805194805 2	0.00064935064935065 51	
F1	0.610830704256272 5	0.566647383405009 2	0.603510076270706 4	-0.0073206279855659 36	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.763636363636363 6	0.733116883116883 1	0.747402597402597 3	-0.016233766233766246	
F1	0.639985235463502 5	0.585955320351279 1	0.611505912294937 3	-0.028479323168565062	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.754545454545454 5	0.7	0.734415584415584 4	-0.020129870129870144	
F1	0.599825946338941 6	0.442024602279676 94	0.565305068241960 1	-0.03452087809698141	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.756493506493506 4	0.724675324675324 6	0.758441558441558 4	0.00194805194805194 3	
F1	0.573368073405794 2	0.424971118952339 6	0.560496913964366 3	-0.012871159441427998	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	0.7454545454545454	0.7558441558441558	0.00649350649350651	
F1	0.6076604413342895	0.5871339176275929	0.6165215801730122	0.00886113883872265	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7487012987012986	0.7344155844155844	0.7357142857142858	-0.01298701298701291	
F1	0.6200247073051919	0.5664465598610027	0.594332699584635	-0.02569200772055703	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6928571428571428	0.6474025974025974	0.6753246753246753	-0.01753246753246755	
F1	0.4761039305152438	0.3780612539056706	0.5351507149761445	0.05904678446090086	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7084415584415584	0.6746753246753248	0.698051948051948	-0.01038961038961038	
F1	0.5724929039023422	0.5192479995055355	0.5542967133927943	-0.01819619050954796	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	0.7629870129870129	0.7616883116883117	-0.01038961038961039	
F1	0.6262914445612621	0.5959186913543753	0.6006164331349428	-0.02567501142631926	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	140.0	52.0	20.0	0.0	29.7	0.128	32.0	1.0	1.0
2	3.0	112.0	70.0	0.0	0.0	25.0	0.197	25.0	0.0	1.0
3	4.0	125.0	72.0	30.0	179.0	44.0706639205383 2	0.137	39.0	0.0	1.0
4	3.0	159.0	70.0	22.0	0.0	20.0703431127114 24	0.94675164754032 18	22.0	1.0	1.0
5	9.0	175.0	88.0	42.0	128.0	34.0	0.58338195485957 73	41.0	1.0	1.0
6	4.0	92.0	68.0	0.0	18.0	22.7636325734665 9	0.238	23.0	0.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	83.0	50.0	15.0	0.0	24.5875315655627 28	0.25243828398344 52	32.0	0.0	2.0
2	4.0	122.0	76.0	23.0	0.0	35.8	0.22813668277403 94	46.0	1.0	2.0
3	5.0	122.0	72.0	11.0	0.0	38.7	0.25207761907774 02	30.0	0.0	2.0
4	7.0	126.0	78.0	0.0	58.0	35.5	0.25139294043983 02	53.0	0.0	2.0
5	4.0	92.0	72.0	0.0	120.0	24.2	0.26186684108695 25	31.0	0.0	2.0
6	3.0	143.0	78.0	0.0	0.0	24.2	0.56031972858035 86	51.0	1.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	125.0	68.0	39.0	91.0	35.0	0.67554793257563	27.0	0.0	3.0
2	1.0	119.0	76.0	0.0	90.0	24.186499816742	0.2609248441736267	24.0	1.0	3.0
3	4.0	189.0	68.0	23.0	0.0	29.0	0.189	45.0	0.0	3.0
4	1.0	123.0	72.0	32.0	125.0	24.8	0.237	28.0	1.0	3.0
5	10.0	95.0	62.0	39.0	0.0	28.9	0.189	39.0	0.0	3.0
6	2.0	110.0	78.0	26.0	180.0	34.6	0.9676810820105874	29.0	0.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	9.0	164.0	79.0	0.0	105.0	25.4	0.284	50.0	0.0	4.0
2	4.0	123.0	70.0	23.0	0.0	24.122471434476644	0.17356632142106126	23.0	1.0	4.0
3	7.0	140.0	78.0	0.0	0.0	35.7	0.23659136694547578	23.0	1.0	4.0
4	5.0	108.0	88.0	0.0	0.0	32.645773763492 45	0.197	41.0	0.0	4.0
5	0.0	12.0	60.0	15.0	0.0	31.6	0.299	22.0	0.0	4.0
6	13.0	171.0	74.0	37.0	76.0	29.5	0.24458538827370874	60.0	0.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	154.0	78.0	33.0	64.0	43.1076039220055 24	0.08816912116947 354	32.0	0.0	5.0
2	1.0	173.0	82.0	28.0	49.0	28.5012657501515 53	0.92245275108557 6	50.0	0.0	5.0
3	4.0	139.0	74.0	0.0	0.0	22.6	0.89846068350646 09	45.0	0.0	5.0
4	4.0	168.0	60.0	50.0	477.0	42.7	1.28251108960383 9	23.0	1.0	5.0
5	2.0	170.0	82.0	0.0	0.0	32.3	0.30311317343198 34	35.0	1.0	5.0
6	1.0	108.0	0.0	27.0	105.0	32.8	1.02314557515042 64	32.0	0.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	10.0	96.0	72.0	26.0	0.0	30.4	0.51526021344180 46	54.0	1.0	6.0
2	4.0	99.0	72.0	0.0	0.0	25.7980095243858 2	0.153	23.0	0.0	6.0
3	2.0	97.0	60.0	0.0	0.0	26.4	0.258	29.0	0.0	6.0
4	5.0	91.0	90.0	37.0	0.0	43.2324847454792 05	0.254	25.0	0.0	6.0
5	13.0	99.0	72.0	41.0	88.0	23.8992514590239 37	0.49465741149688 447	39.0	0.0	6.0
6	3.0	93.0	68.0	0.0	0.0	26.1731937822452 64	0.141	21.0	0.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	112.0	82.0	0.0	0.0	28.7	0.20034846154157396	25.0	1.0	7.0
2	11.0	119.0	72.0	42.0	215.0	39.992003200105001	1.6351232174187795	52.0	1.0	7.0
3	5.0	126.0	74.0	35.0	0.0	41.5	0.2366734198548699	56.0	0.0	7.0
4	4.0	95.0	78.0	0.0	0.0	34.6	0.2646239972869208	30.0	0.0	7.0
5	8.0	136.0	72.0	26.0	0.0	35.54114050630519	0.234	65.0	0.0	7.0
6	7.0	108.0	80.0	0.0	0.0	43.443719424681966	0.25153372273427993	40.0	0.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	95.0	84.0	18.0	0.0	34.512509217512054	0.43211465385904513	24.0	0.0	8.0
2	0.0	167.0	78.0	20.0	200.0	30.742039456564101	0.5156973614825091	23.0	0.0	8.0
3	2.0	179.0	62.0	32.0	478.0	32.4	0.14195884584420587	23.0	0.0	8.0
4	9.0	112.0	80.0	39.0	0.0	33.7	0.5154361046386031	38.0	1.0	8.0
5	3.0	131.0	74.0	0.0	0.0	23.7	0.159	69.0	0.0	8.0
6	6.0	149.0	88.0	35.0	115.0	40.558861703324009	0.245	25.0	1.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	129.0	74.0	37.0	180.0	29.9	0.7190983828422924	48.0	0.0	9.0
2	1.0	125.0	82.0	31.0	0.0	44.0	0.9131726430927944	53.0	0.0	9.0
3	1.0	122.0	48.0	27.0	270.0	24.337734704833437	0.6667493463295721	41.0	0.0	9.0
4	10.0	132.0	88.0	33.0	0.0	38.41035274514449	0.13382667517449293	33.0	0.0	9.0
5	7.0	125.0	0.0	0.0	0.0	38.091655592221118	0.7366259834171492	45.0	0.0	9.0
6	0.0	89.0	0.0	27.0	0.0	28.5	0.4842372695056929	25.0	0.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	137.0	74.0	33.0	0.0	35.3	0.4818907677771364	37.0	1.0	10.0
2	8.0	96.0	56.0	36.0	120.0	32.8	0.19	26.0	0.0	10.0
3	7.0	109.0	90.0	18.0	0.0	39.9	0.09417619307231205	27.0	0.0	10.0
4	11.0	112.0	60.0	0.0	67.0	36.10349535420492	0.6727128507940925	27.0	0.0	10.0
5	5.0	114.0	64.0	11.0	0.0	33.6	0.2213742070725515	32.0	0.0	10.0
6	12.0	85.0	62.0	0.0	0.0	24.2	0.5874101987992021	67.0	0.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan